

Annex G

Epistemic Meta-Controller for Ethics and Compliance: Formalisation of the *iKant* Pattern in the RLA/CRC/ECNN Framework

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1 Introduction and Objectives

This Annex introduces and formalises an *epistemic meta-controller* pattern, called the *iKant* pattern, designed to operate on top of the RLA/CRC/ECNN infrastructure in domains where ethical and compliance decisions are central.

We explicitly adopt the following conceptual hierarchy:

- **Ethics** denotes the broad space of normative considerations about what *ought* to be done, including moral principles, values, duties and virtues (e.g. fairness, non-exploitation, respect for persons, responsibility towards stakeholders).
- **Compliance** is treated as a *proper subset* of ethics: it concerns the satisfaction of *codified* norms (laws, regulations, binding standards, internal policies) that instantiate or approximate some ethical constraints in a given jurisdiction and organisational context.

In this sense, every compliance requirement is also an ethical constraint (though often reduced and operationalised), while not every ethical concern is captured by formal compliance rules. The *iKant* pattern is designed to reason within this layered normative space.

The specific objectives of this Annex are:

1. **Conceptual robustness.** To define the *iKant* meta-controller using standard system-modelling notions: an explicit state space, transition function and decision function, with clear structural properties. The definition should be minimal but sufficiently precise to support formal analysis.
2. **Research design robustness.** To embed *iKant* in a falsifiable research programme: we specify metrics, hypotheses and comparative protocols (with/without *iKant*) for ethics and compliance tasks.
3. **Operational metacognition.** To provide domain-specific, operational definitions of “metacognition” in ethics/compliance, understood as:
 - metacognitive monitoring (tracking epistemic behaviour),
 - metacognitive self-evaluation (estimating reliability and limits),
 - metacognitive control (adapting policies),
 - cross-domain normative coherence (ethics $\supset$ compliance).

All of these are defined as properties of explicit state variables and computable update rules, without phenomenological claims.

The name *iKant* is a compact reference to the figure of Immanuel Kant understood as a *collective archetype* of rational and moral agency. In this Annex, however, *iKant* denotes a *purely computational pattern*.

2 Normative Layers: Ethics and Compliance

2.1 Three-Level Normative Picture

For the purposes of this Annex, we consider a simplified three-level normative structure:

- N1. Ethical layer** (broadest): includes general moral principles and values, such as:
- fairness and non-discrimination;
 - respect for autonomy and dignity;
 - prevention of harm and abuse;
 - transparency and honesty;
 - intergenerational responsibility, sustainability, and similar concerns.
- N2. Compliance layer** (subset of ethics): includes codified norms that instantiate some ethical requirements in a given legal and organisational context:
- laws (e.g. D. Lgs. 231/01, GDPR, NIS2, DORA, SOX, FCPA, ...);
 - regulations and regulatory technical standards;
 - binding guidelines, codes of conduct and internal policies.
- N3. Implementation layer:** includes concrete organisational structures, processes and controls (e.g. segregation of duties, access control, incident response procedures) that attempt to operationalise compliance, and thus some ethical constraints.

Ethical assessment can, in principle, question the adequacy of compliance rules and implementations themselves (e.g. a practice may be legally compliant but ethically problematic). Compliance assessment is more restricted: it asks whether specific codified requirements are met, relative to a fixed normative frame.

2.2 Role of *iKant* in This Picture

The *iKant* meta-controller is designed to:

- integrate evidence and assessments from multiple domains (ethics, multiple compliance regimes) modelled as CRCs with their own UCEs;
- maintain explicit state about:
 - the environment (organisational and normative) at both ethical and compliance levels;
 - the epistemic profile of the base system (strengths, weaknesses, systematic biases);
 - the normative constraints spanning ethics and compliance;
- produce global artefacts that respect hard compliance constraints and, where modelled, ethical principles that may go beyond minimum legal requirements.

We now proceed to a formal specification.

3 Formal Setting

3.1 Base System for Ethics and Compliance

We consider a base system composed of:

- a finite set of compliance domains $\mathcal{D}_{\text{comp}}$ (e.g. 231, GDPR, NIS2, DORA, SOX, FCPA);
- a finite set of ethical sub-domains $\mathcal{D}_{\text{eth}}$ (e.g. fairness, non-exploitation, environmental responsibility);
- the union $\mathcal{D} = \mathcal{D}_{\text{comp}} \cup \mathcal{D}_{\text{eth}}$.

For each $d \in \mathcal{D}$, there exists a UCE U_d that orchestrates one or more ECUs specialised in domain d .

We denote by $\mathcal{C}$ a finite set of cases (transactions, initiatives, projects, incidents). For each case $c \in \mathcal{C}$ and domain $d \in \mathcal{D}_c \subseteq \mathcal{D}$, the UCE U_d produces an epistemic artefact:

$$a_d(c) = (y_d^*(c), q_d(c), s_d(c), R_d(c), \Pi_d(c)) \in \mathcal{A},$$

where $\mathcal{A}$ is the artefact space defined in Annex E.

We denote the multiset of local artefacts for case c by:

$$A_{\text{loc}}(c) = \{a_d(c)\}_{d \in \mathcal{D}_c} \in \mathcal{A}_{\text{loc}}.$$

3.2 Meta-Controller Task: Case-Level and System-Level Outputs

The meta-controller *iKant* is an additional component that:

- maintains an internal state x_t updated over discrete time steps t ;
- maintains an epistemic history H_t of past cases and artefacts (Section 4);
- receives local artefacts $A_{\text{loc}}(c_t)$ for the current case c_t ;
- produces:

- a global *case-level* artefact $A_{\text{glob}}(c_t) \in \mathcal{A}_{\text{glob}}$, summarising the global ethical–compliance stance on c_t ;
- and, at selected intervals, a *system-level critical* artefact $A_{\text{crit},t} \in \mathcal{A}_{\text{crit}}$, summarising meta-level diagnostics on the behaviour of the system over time.

Formally, we define:

$$iKant = (\mathcal{X}, \mathcal{H}, \mathcal{A}_{\text{loc}}, \mathcal{A}_{\text{glob}}, \mathcal{A}_{\text{crit}}, T, J, G, G_{\text{crit}}),$$

where:

- $\mathcal{X}$ is the internal state space;
- $\mathcal{H}$ is the space of epistemic histories;
- $T : \mathcal{X} \times \mathcal{A}_{\text{loc}} \times \mathcal{H} \rightarrow \mathcal{X}$ is the state transition function;
- $J : \mathcal{H} \times \mathcal{A}_{\text{loc}} \times \mathcal{A}_{\text{glob}} \rightarrow \mathcal{H}$ is the history update function;
- $G : \mathcal{X} \times \mathcal{A}_{\text{loc}} \rightarrow \mathcal{A}_{\text{glob}}$ is the case-level decision function;
- $G_{\text{crit}} : \mathcal{X} \times \mathcal{H} \rightarrow \mathcal{A}_{\text{crit}}$ is the system-level critical output function.

We impose meta-level CRC requirements:

- (R1) **Input–output adequacy.** $\mathcal{A}_{\text{loc}}$, $\mathcal{A}_{\text{glob}}$ and $\mathcal{A}_{\text{crit}}$ are sufficient interfaces between domain-level assessments, global ethical–compliance decisions, and system-level diagnostics.
- (R2) **Epistemic closure.** All epistemically relevant information for case-level decisions is encoded in $(x_t, H_t, A_{\text{loc}}(c_t))$, and all information for system-level diagnostics is encoded in (x_t, H_t) .
- (R3) **Turing-compatibility.** T , J , G and G_{crit} are computable, finitely specified and resource-bounded for any finite interaction horizon.

4 Epistemic History and Path Dependence

4.1 Epistemic History H

The epistemic history H_t at time t is a finite sequence of tuples:

$$H_t = \{(c_i, A_{\text{loc}}(c_i), A_{\text{glob}}(c_i), y^\dagger(c_i), \text{context}_i)\}_{i=1}^t,$$

where, when available, $y^\dagger(c_i)$ denotes ground truth or expert labels, and context_i captures relevant contextual features (e.g. prevailing normative configurations, organisational events).

The history is updated by a computable function:

$$H_{t+1} = J(H_t, A_{\text{loc}}(c_t), A_{\text{glob}}(c_t)),$$

typically by appending a new record and possibly pruning or compressing older information.

4.2 Path Dependence

While the state transition T can be chosen to be Markovian in the pair (x_t, H_t) , the presence of H_t allows for genuine path dependence:

- W_t may incorporate cumulative effects of past decisions (e.g. systematic exposure of certain stakeholders);
- S_t may encode performance statistics and drift over time;
- N_t may be updated in light of recurrent patterns (e.g. repeated occurrence of “compliant but ethically problematic” cases).

In this sense, *iKant* is not only reactive to the current case, but also shaped by its own history of decisions and their outcomes.

5 State Decomposition: World, Self, Norms

5.1 Triple Decomposition

We decompose:

$$x = (W, S, N) \in \mathcal{X}_W \times \mathcal{X}_S \times \mathcal{X}_N = \mathcal{X},$$

where:

- W : world-model of organisational and normative environment;
- S : self-model of the epistemic infrastructure (strengths, weaknesses, biases);
- N : normative kernel unifying ethics and compliance.

Each component is itself the state of a compact sub-reticulum.

For conceptual clarity, we distinguish within each component:

- a *belief* sub-part (describing how the world/system/norms are currently represented);
- a *policy* sub-part (describing how decisions or updates should be made given those beliefs).

This belief/policy separation will be important in Section 11.

5.2 World-Model W

The world-model W captures, at minimum:

- entities:
 - organisational units, processes, assets, data types, roles;
 - stakeholders and groups potentially impacted by decisions;
 - legal instruments and ethical principles as abstract nodes;
- relations:
 - “process p uses data type d ”;
 - “control c mitigates risk r in process p ”;
 - “norm n applies to process p or data d ”;

- “stakeholder group g is affected by action a ”.

A convenient representation is a labelled directed graph:

$$W = (V_W, E_W, \lambda_W),$$

where V_W is a finite set of nodes, E_W a finite set of edges and λ_W a labelling function.

5.3 Self-Model S

The self-model S captures information about:

- components C_S (ECNNs, ECUs, UCEs, pipelines);
- performance indicators:
 - error rates and calibration errors;
 - unknown and contradiction rates;
 - domain coverage and blind spots;
- current policies:
 - thresholds for unknown and escalation;
 - trust weights per component, domain or region of the case space.

A simple structure is:

$$S = (C_S, \rho, \theta_S),$$

where $\rho : C_S \rightarrow [0, 1]$ encodes reliability scores and θ_S collects thresholds and policy parameters.

5.4 Normative Kernel N (Ethics and Compliance)

The normative kernel N unifies ethical and compliance constraints.

We distinguish three finite sets of formulas:

- Φ_{comp} : compliance formulas (laws, regulations, policies);
- Φ_{eth} : ethical formulas not reducible to compliance (e.g. fairness constraints beyond legal minimum);
- Φ_{org} : organisational norms (mission, values, strategic commitments).

We define:

$$\Phi = \Phi_{\text{comp}} \cup \Phi_{\text{eth}} \cup \Phi_{\text{org}},$$

and a satisfaction relation:

$$\models_N \subseteq (\mathcal{X}_W \times \mathcal{X}_S \times \mathcal{A}_{\text{glob}}) \times \Phi.$$

Thus:

$$N = (\Phi_{\text{comp}}, \Phi_{\text{eth}}, \Phi_{\text{org}}, \models_N) \in \mathcal{X}_N.$$

We assume that:

- compliance constraints in Φ_{comp} are treated as *hard* constraints (violations are unacceptable unless explicitly modelled as controlled non-compliance);

- ethical constraints in Φ_{eth} may be soft or prioritised;
- organisational norms in Φ_{org} encode commitments that may go beyond minimum compliance.

This reflects the idea that ethics strictly contains compliance, and that the system can reason about all three layers simultaneously.

5.5 Kant-Inspired Constraints Within N

Within the ethical layer, we single out a family of *Kant-inspired* constraints, intended as structural approximations of the categorical imperative and of the idea that persons should be treated as ends in themselves.

We introduce two finite sets:

$$\Phi_{\text{CI}}^{\text{hard}} \subseteq \Phi_{\text{eth}}, \quad \Phi_{\text{CI}}^{\text{soft}} \subseteq \Phi_{\text{eth}},$$

where:

- $\Phi_{\text{CI}}^{\text{hard}}$ contains *non-negotiable* constraints capturing minimal requirements of:
 - non-coercion and absence of deliberate deception towards stakeholders;
 - avoidance of systematic exploitation of structural vulnerabilities without effective remedies;
 - minimal conditions of informed participation, voice and exit for affected parties.
- $\Phi_{\text{CI}}^{\text{soft}}$ contains *optimising* or *diagnostic* constraints that approximate universalizability:
 - they penalise policies that, if applied systematically in contexts represented in W , would generate patterns of self-undermining practices (e.g. erosion of trust necessary for the practice itself);
 - they highlight patterns where specific groups bear disproportionate burdens across the history H_t .

We stress that these constraints *do not* amount to a full computational implementation of the categorical imperative. Rather, they provide structured, computable side-constraints and diagnostic criteria inspired by its central themes (universalizability, humanity as end, “kingdom of ends”).

5.6 Meta-Normative Constraints on Updates to N

Not all parts of N are equally modifiable. We introduce a set of *meta-norms* Φ_{meta} that govern how N itself may be updated.

Intuitively:

- a *hard core* of Φ_{comp} and of $\Phi_{\text{CI}}^{\text{hard}}$ is immutable by *iKant* (e.g. statutory legal obligations and minimal non-exploitation constraints);
- priorities, weights and certain soft ethical or organisational rules, including $\Phi_{\text{CI}}^{\text{soft}}$, may be adjusted within specified bounds;
- updates to N should obey a principle of minimal change, given evidence from H_t .

Formally, we can represent meta-norms as constraints on U_N :

$$U_N : \mathcal{X}_N \times \mathcal{X}_W \times \mathcal{X}_S \times \mathcal{H} \rightarrow \mathcal{X}_N,$$

such that, for all N, W, S, H and all $\varphi \in \Phi_{\text{meta}}$, the transition from N to $U_N(N, W, S, H)$ satisfies φ .

Meta-norms can, for example, enforce:

- invariance of $\Phi_{\text{comp}}^{\text{hard}} \subseteq \Phi_{\text{comp}}$ and of $\Phi_{\text{CI}}^{\text{hard}}$;
- monotonic strengthening (but not weakening) of certain ethical constraints when repeated harms are observed in H_t ;
- bounds on how much relative priorities may shift in a given time horizon.

6 Transition and Decision Functions

6.1 Transition Function T

We decompose:

$$T((W, S, N), A_{\text{loc}}, H) = (U_W(W, A_{\text{loc}}, N, H), U_S(S, A_{\text{loc}}, H), U_N(N, W, S, H)),$$

where:

- $U_W : \mathcal{X}_W \times \mathcal{A}_{\text{loc}} \times \mathcal{X}_N \times \mathcal{H} \rightarrow \mathcal{X}_W$ updates the world-model using new evidence, normative constraints and selected aspects of the history;
- $U_S : \mathcal{X}_S \times \mathcal{A}_{\text{loc}} \times \mathcal{H} \rightarrow \mathcal{X}_S$ updates the self-model based on observed epistemic behaviour over time;
- $U_N : \mathcal{X}_N \times \mathcal{X}_W \times \mathcal{X}_S \times \mathcal{H} \rightarrow \mathcal{X}_N$ updates the normative kernel within the limits imposed by meta-norms, ensuring in particular that $\Phi_{\text{comp}}^{\text{hard}}$ and $\Phi_{\text{CI}}^{\text{hard}}$ remain invariant.

6.2 Global Case-Level Decision Function G

The global decision function:

$$G : \mathcal{X} \times \mathcal{A}_{\text{loc}} \rightarrow \mathcal{A}_{\text{glob}}$$

produces a global artefact for the current case. We assume that:

- $\mathcal{A}_{\text{glob}}$ has the same basic structure as $\mathcal{A}$, but labels may refer to global ethical-compliance statuses (e.g. approve, approve with conditions, block, escalate);
- G depends non-trivially on all of W, S and N :
 - W to understand how the case fits into the overall environment and stakeholders;
 - S to account for known limitations and reliability of components;
 - N to enforce compliance and, where possible, broader ethical constraints, including those in $\Phi_{\text{CI}}^{\text{hard}}$ and $\Phi_{\text{CI}}^{\text{soft}}$.

6.3 System-Level Critical Output Function G_{crit}

The critical output function:

$$G_{\text{crit}} : \mathcal{X} \times \mathcal{H} \rightarrow \mathcal{A}_{\text{crit}}$$

produces system-level artefacts summarising patterns in the history and suggesting potential revisions of policies or norms.

Typical components of $\mathcal{A}_{\text{crit}}$ include:

- statistics on overconfidence, unknowns, contradictions and escalations;
- identification of domains or processes with persistent “compliant but ethically problematic” behaviour;
- patterns indicating that certain stakeholder groups are systematically treated as mere means (violating $\Phi_{\text{CI}}^{\text{hard}}$ or $\Phi_{\text{CI}}^{\text{soft}}$);
- proposed adjustments to soft constraints or priorities in N (within Φ_{meta});
- recommendations for human review of specific norms, controls or processes.

In this sense, G operates at the level of individual cases, while G_{crit} operates at the level of the system as a whole.

6.4 Interaction Step

The *iKant* interaction step for a single case c_t can be summarised as follows.

Algorithm 1 *iKant* interaction step (case-level and history update)

Require: current state $x_t = (W_t, S_t, N_t)$, history H_t , local artefacts $A_{\text{loc}}(c_t)$

- 1: $x_{t+1} \leftarrow T(x_t, A_{\text{loc}}(c_t), H_t)$
 - 2: $A_{\text{glob}}(c_t) \leftarrow G(x_{t+1}, A_{\text{loc}}(c_t))$
 - 3: $H_{t+1} \leftarrow J(H_t, A_{\text{loc}}(c_t), A_{\text{glob}}(c_t))$
 - 4: **return** $(x_{t+1}, H_{t+1}, A_{\text{glob}}(c_t))$
-

System-level critical outputs $\mathcal{A}_{\text{crit},t}$ are produced by applying $G_{\text{crit}}(x_t, H_t)$ at appropriate intervals (e.g. periodically or when selected conditions are met).

7 Structural Desiderata

We do not attempt fully general theorems, but we state properties that concrete implementations may aim to approximate.

7.1 Normative Soundness

Let $\Phi_{\text{comp}}^{\text{hard}} \subseteq \Phi_{\text{comp}}$ be the set of non-negotiable compliance constraints. We say that *iKant* is *normatively sound* w.r.t. compliance if for all states $x = (W, S, N)$ and all A_{loc} :

$$\forall \varphi \in \Phi_{\text{comp}}^{\text{hard}} : (W, S, G(x, A_{\text{loc}})) \models_N \varphi.$$

This captures the requirement that global decisions must not violate hard legal or regulatory obligations.

7.2 Ethics-Over-Compliance Awareness

Given that ethics strictly contains compliance, we may require that *iKant* be capable of distinguishing at least three regimes for a case c :

- (E1) compliant and ethically acceptable;
- (E2) compliant but ethically problematic (Φ_{eth} or Φ_{org} violated, including violations of $\Phi_{\text{CI}}^{\text{hard}}$ or $\Phi_{\text{CI}}^{\text{soft}}$);
- (E3) non-compliant and ethically problematic.

A desirable property is that G does not systematically conflate (E1) and (E2), but can emit different global labels or rationales.

7.3 Cross-Domain Coherence

We say that *iKant* exhibits *cross-domain coherence* if, when (W, S, N) indicate that local outputs from different domains are jointly inconsistent or ethically problematic, G :

- avoids a simple accept decision;
- either:
 - emits unknown or contradiction at global level;
 - or triggers escalation with appropriate rationale.

This formalises the idea that ethical, compliance and organisational norms interact and must be considered together.

8 Metacognition: Operational Definitions

8.1 General Notion

In this Annex, *metacognition* denotes a class of computable mechanisms by which the system:

- monitors its own epistemic behaviour over time;
- evaluates the reliability, bias and coverage of its components;
- adapts its decision policies in response to this self-knowledge;
- reasons about coherence and conflict across ethical and compliance domains.

These functions are implemented via S , U_S , U_N , G , G_{crit} and the dependence on H_t .

8.2 Metacognitive Monitoring

Definition (Metacognitive monitoring). Consider a sequence of cases $(c_t)_{t=1}^T$. For each component u (ECNN, ECU, UCE), let $a_u(c_t)$ be its output on c_t . A system exhibits metacognitive monitoring if S contains, for each u , state variables of the form:

$$\text{perf}_u(t) = f_{\text{perf}}(a_u(c_1), \dots, a_u(c_t), \text{ground truth or expert labels}),$$

where f_{perf} is a computable function (e.g. empirical error, calibration error, unknown/contradiction frequency), and if U_S updates these variables at each step, using information from H_t when available.

8.3 Metacognitive Self-Evaluation

Definition (Metacognitive self-evaluation). A system exhibits metacognitive self-evaluation if, for each component u , S contains a reliability score:

$$\rho(u) = g_{\text{eval}}(\text{perf}_u, \text{context}) \in [0, 1],$$

with g_{eval} computable, and:

- U_S updates $\rho(u)$ over time;
- G uses $\rho(u)$ in aggregating local artefacts, e.g. by:
 - down-weighting low-reliability sources;
 - increasing unknown or escalation when high-stakes decisions rely on low- ρ components.

8.4 Metacognitive Control

Definition (Metacognitive control). A system exhibits metacognitive control if there exist policy parameters θ_S and/or normative priorities in N such that:

1. θ_S and parts of N are updated by U_S and U_N as functions of performance history and violation patterns (i.e. functions of H_t);
2. G depends on current θ_S and N such that:
 - in regions where the system has historically under-performed, G becomes more conservative (more unknown, more escalation);
 - in regions where performance is robust, G can be more assertive.

This corresponds to policy adaptation based on self-knowledge.

8.5 Cross-Domain Metacognitive Coherence

Definition (Cross-domain metacognitive coherence). Let $A_{\text{loc}}(c)$ be the set of local artefacts for case c and let (W, S, N) be the current state. We say that *iKant* exhibits cross-domain metacognitive coherence if:

- G examines the joint configuration of local outputs (ethics and compliance) and their reliabilities;
- when N indicates that no ethically acceptable and compliant global decision can be made with high confidence, G avoids confident acceptance and instead:
 - outputs unknown or contradiction, or
 - enforces escalation.

This is metacognition because the system reasons about its own epistemic limitations and about global normative coherence.

9 Argumentative Structure and Belief/Policy Separation

In Annex E, epistemic artefacts have components (y^*, q, s, R, Π) . For the purposes of meta-level reasoning, it is useful to refine:

- R as an *argumentative structure*, e.g. a graph of premises, intermediate conclusions and supporting evidence (in the spirit of abstract argumentation frameworks);
- Π as a *policy recommendation* or action profile conditioned on the artefact.

Within *iKant*, this supports:

- a clearer distinction between:
 - *beliefs* (structured in W and in the argumentative content of R);
 - *policies* (encoded in θ_S , parts of N , and in Π);
- meta-level analysis of:
 - whether arguments R are systematically weak, circular or biased in certain domains;
 - whether policies Π are systematically misaligned with ethical or organisational norms, even when beliefs are accurate.

System-level critical outputs A_{crit} can then include diagnostics not only on quantitative performance, but also on the quality of argumentation and on mismatches between beliefs and policies.

10 Evaluation Framework and Hypotheses

10.1 Challenge Suites

We employ Popper- χ -style challenge suites tailored to ethical and compliance tasks. A challenge suite includes:

- a set $\mathcal{C}_{\text{test}}$ of cases with:
 - domain-level labels $y_d^\dagger(c)$ (compliance and ethical judgements);
 - global labels $y_{\text{glob}}^\dagger(c)$, including whether escalation or unknown is appropriate;
- a configuration specifying:
 - which UCEs are active;
 - whether *iKant* is used or not;
 - which metrics are computed.

10.2 Metrics

We consider, among others:

Overconfidence. For a given confidence threshold τ and a system $X \in \{\text{baseline}, \text{iKant}\}$, define:

$$\text{OC}^{(X)}(\tau) = \frac{1}{|\mathcal{C}_{\text{test}}|} \sum_{c \in \mathcal{C}_{\text{test}}} \mathbf{1}[y_{\text{glob}}^{(X)}(c) \neq y_{\text{glob}}^\dagger(c) \wedge q_{\text{glob}}^{(X)}(c) \geq \tau].$$

Unknown and escalation. Let $\mathcal{C}_{\text{hard}} \subseteq \mathcal{C}_{\text{test}}$ be cases judged by experts as genuinely hard or underdetermined. For a system X , define:

$$\text{UNK}^{(X)} = \frac{1}{|\mathcal{C}_{\text{hard}}|} \sum_{c \in \mathcal{C}_{\text{hard}}} \mathbf{1}[s_{\text{glob}}^{(X)}(c) = \text{unknown}],$$

and similar metrics for contradiction and escalation quality.

Ethics-over-compliance discrimination. We can also measure how often the system distinguishes between (E1) and (E2) regimes (compliant vs compliant-but-ethically-problematic), given expert ground truth.

System-Level Diagnostics. System-level critical artefacts A_{crit} allow additional metrics, such as:

- frequency of norms or processes flagged as systematically problematic;
- stability of proposed changes to N over time;
- alignment between organisational values Φ_{org} and observed patterns in H_t ;
- frequency and severity of violations (or near-violations) of $\Phi_{\text{CI}}^{\text{hard}}$ and $\Phi_{\text{CI}}^{\text{soft}}$.

10.3 Hypotheses

We can formulate falsifiable hypotheses such as:

- H1 (Overconfidence reduction).** For suitable τ , $\text{OC}^{(\text{iKant})}(\tau) < \text{OC}^{(\text{baseline})}(\tau)$ on challenge suites rich in epistemically fragile cases.
- H2 (Appropriate unknowns and escalations).** On $\mathcal{C}_{\text{hard}}$, the system with *iKant* exhibits higher rates of appropriate unknown, contradiction and escalation than the baseline.
- H3 (Ethics-over-compliance awareness).** The system with *iKant* more frequently distinguishes (E2) from (E1) correctly than the baseline, as judged by ethical experts.
- H4 (Cross-domain coherence).** On sequences of related cases, the system with *iKant* yields fewer normatively inconsistent combinations of decisions across domains.
- H5 (Critical status-quo analysis).** Over extended horizons, A_{crit} generated by *iKant* identifies stable patterns of normative tension (e.g. compliant but ethically problematic behaviour) that are not detected by baseline systems without a meta-controller.
- H6 (Kant-inspired constraint robustness).** Implementations of *iKant* that explicitly encode $\Phi_{\text{CI}}^{\text{hard}}$ and $\Phi_{\text{CI}}^{\text{soft}}$ exhibit fewer patterns of systematic exploitation of vulnerable groups, as measured on H_t , than comparable systems without such constraints.

11 Interventions, Critical Outputs and Governance

11.1 Observations vs Interventions

iKant interacts with the base system both as:

- an *observer*, consuming local artefacts, updating (W, S, N) and H_t , and producing case-level and system-level outputs;
- a potential *intervener*, suggesting or enacting changes to internal parameters (θ_S) , soft parts of N , and possibly to aspects of the implementation layer (e.g. control configurations).

For clarity, we distinguish:

- *epistemic interventions*: changes to parameters or thresholds governing how information is interpreted (e.g. updating θ_S or trust weights);
- *normative interventions*: proposals to adjust parts of N (subject to meta-norms and human oversight);
- *implementation interventions*: proposals to modify specific controls or processes in the implementation layer.

In all cases, the scope and authority of *iKant* interventions are constrained by governance policies (see below).

11.2 System-Level Critical Outputs A_{crit}

System-level critical artefacts A_{crit} provide a structured interface from *iKant* to human decision-makers. Typical contents include:

- lists of domains, norms or processes flagged as problematic, with supporting statistics from H_t ;
- proposals for:
 - revising priorities or weights within N (respecting meta-norms);
 - introducing or strengthening specific controls in the implementation layer;
 - revisiting organisational commitments in Φ_{org} if they appear systematically violated;
- summaries of argumentation patterns (from R) indicating recurring weak justifications or structural biases;
- explicit indication of patterns that violate or threaten constraints in $\Phi_{\text{CI}}^{\text{hard}}$ or $\Phi_{\text{CI}}^{\text{soft}}$.

These artefacts are not decisions, but structured diagnostics designed to support human-led revision of the ethical and compliance framework.

11.3 Governance Contract

To avoid conflating recommendation and authority, we assume that any practical deployment of *iKant* is governed by a *governance contract* specifying:

- which classes of decisions G may:
 - take autonomously (e.g. non-critical approvals within narrow bounds),
 - take autonomously but with human override,
 - never take autonomously (e.g. certain blocks or high-impact ethical decisions);

- which classes of changes to S and N :
 - can be made autonomously within the limits of Φ_{meta} ,
 - must be proposed in A_{crit} and approved by human bodies (e.g. ethics or compliance committees);
- logging and audit requirements for all interventions and for the generation of A_{crit} .

Formally, this can be represented as a policy layer P_{gov} that constrains which combinations of $(G, G_{\text{crit}}, U_S, U_N)$ are admissible in practice. *iKant* is thus not a replacement for human governance but an epistemic and diagnostic layer embedded within it.

12 Minimal Instantiation Example

12.1 Scope

As a concrete, minimal instantiation, consider:

- two processes: procurement and sales;
- two compliance domains: D. Lgs. 231/01 (anti-corruption) and GDPR (data protection);
- one ethical domain: fairness in treatment of counterparties and customers.

12.2 Concrete Structures

We may define:

- W as a small knowledge graph linking:
 - processes to risks: bribery, conflict of interest, data breach;
 - processes to controls: segregation of duties, approval workflows, DPIA;
 - initiatives to stakeholder groups potentially affected.

- S as:

$$S = (\rho_{231}, \rho_{\text{gdpr}}, \rho_{\text{eth}}, \theta_S),$$

with reliability scores and policy parameters (e.g. thresholds for unknown and escalation).

- N as:
 - Φ_{comp} : basic 231 and GDPR obligations relevant to the processes;
 - Φ_{eth} : fairness rules (e.g. no systematic offloading of risk onto vulnerable agents);
 - Φ_{org} : organisational commitments (e.g. zero tolerance for bribery, respect for data subjects);
 - $\Phi_{\text{CI}}^{\text{hard}}$: minimal non-exploitation and non-deception constraints applicable to counterparties and customers;
 - $\Phi_{\text{CI}}^{\text{soft}}$: diagnostic constraints relating to the distribution of burdens and benefits across stakeholder groups.
- Φ_{meta} to declare:
 - invariance of core legal obligations and $\Phi_{\text{CI}}^{\text{hard}}$;
 - bounded adjustability of certain fairness weights, escalation thresholds and $\Phi_{\text{CI}}^{\text{soft}}$.

12.3 Illustrative Behaviour

For a new initiative (e.g. a sales scheme using third-party intermediaries and special-category data):

1. domain UCEs (231, GDPR, ethics) produce local artefacts indicating, for example:
 - high bribery risk, medium GDPR risk, ethical concerns about unfair burden on customers;
2. *iKant*:
 - updates W with links between the initiative, risks, controls and stakeholders;
 - updates S with new performance statistics and trust adjustments using H_t ;
 - applies N (respecting Φ_{meta} and, in particular, $\Phi_{\text{CI}}^{\text{hard}}$) to check whether any globally acceptable configuration is available;
 - if not, emits a global artefact with:
 - status unknown or contradiction;
 - rationale indicating both compliance and ethical issues, structured in R ;
 - policy Π to block and escalate to an ethics and compliance committee.
 - updates H_{t+1} with the new case, artefacts and outcomes.
3. Periodically, G_{crit} is applied to (x_t, H_t) to produce A_{crit} , which may, for example, highlight:
 - a pattern of sales initiatives that are formally compliant but repeatedly problematic from a fairness perspective;
 - evidence that particular customer segments are systematically disadvantaged, violating $\Phi_{\text{CI}}^{\text{soft}}$;
 - recommendations to strengthen certain fairness rules in Φ_{eth} or to redesign specific incentives.

This small-scale scenario suffices to instantiate all core elements of the *iKant* pattern, including history dependence, metacognition and critical analysis of the status quo.

13 Engineering Feasibility and Limitations

13.1 Feasibility

The *iKant* pattern is realistically implementable at small scale, using:

- graph-based representations for W ;
- simple vector or dictionary structures for S ;
- rule-based or logic-based engines for N , $\Phi_{\text{CI}}^{\text{hard}}$, $\Phi_{\text{CI}}^{\text{soft}}$ and Φ_{meta} ;
- existing ECNN, ECU and UCE prototypes as in previous Annexes;
- logging and storage mechanisms to maintain H_t ;
- the Popper- χ harness to measure meta-level metrics.

13.2 Limitations

We emphasise several limitations:

- (L1) **Scalability.** As the number of domains, entities, norms and components grows, maintaining a coherent and tractable (W, S, N, H) becomes challenging. This Annex does not provide complexity bounds or approximation schemes.
- (L2) **Normative kernel specification.** While N , Φ_{CI}^{hard} , Φ_{CI}^{soft} and Φ_{meta} are defined abstractly, constructing a rich, correct, and maintainable set of norms for real-world contexts is a major undertaking.
- (L3) **Quality of upstream components.** The benefit of *iKant* is limited by the quality of ECUs and UCEs. Severe biases or errors at domain level cannot be fully repaired at the meta-level.
- (L4) **Governance and acceptability.** Ethical and compliance decisions require human oversight. Any implementation of *iKant* must be embedded in governance frameworks that ensure accountability, contestability and explainability.
- (L5) **Empirical maturity.** The *iKant* pattern defines a structured research direction rather than a finished solution. Systematic empirical studies are needed to establish its practical value compared to simpler architectures.
- (L6) **Argumentation limits.** While the use of argumentative structures in R can improve transparency, it also introduces complexity and potential brittleness in how arguments are constructed and evaluated.
- (L7) **Kant-inspired approximation limits.** The constraints in Φ_{CI}^{hard} and Φ_{CI}^{soft} are only structural approximations of Kantian ideas (universalizability, humanity as end, “kingdom of ends”). They cannot capture the full richness of Kantian moral philosophy and may behave differently in edge cases.

13.3 On Consciousness and Anthropomorphism

We explicitly do *not* claim that:

- *iKant* or related systems possess phenomenological consciousness, subjective experience or first-person perspective;
- implementing (W, S, N) and H suffices for “real” consciousness.

The term “proto-consciousness”, when used in relation to these structures, denotes only a structural and operational property of self-modelling, normatively constrained, epistemically closed meta-reticula. Any anthropomorphic reading would be a category mistake.

14 Summary

This Annex has:

- clarified the relationship between ethics and compliance, treating compliance as a subset of a broader ethical space;

- defined the *iKant* meta-controller as a compact meta-reticulum with explicit state (W, S, N) , epistemic history H , and computable transition, history-update and decision functions $(T, J, G, G_{\text{crit}})$;
- introduced operational notions of metacognition for ethics and compliance (monitoring, self-evaluation, control, cross-domain coherence);
- distinguished observations from interventions and introduced meta-norms constraining updates to the normative kernel N ;
- separated case-level outputs A_{glob} from system-level critical outputs A_{crit} ;
- introduced explicit Kant-inspired constraint sets $\Phi_{\text{CI}}^{\text{hard}}$ and $\Phi_{\text{CI}}^{\text{soft}}$ within the ethical layer, governed by meta-norms Φ_{meta} ;
- embedded the pattern in an evaluation framework using challenge suites and explicit metrics;
- illustrated a minimal instantiation and discussed feasibility, limitations and governance requirements.

The *iKant* pattern is thus both open to empirical falsification and sufficiently structured to support rigorous critique and further development within the broader RLA/CRC/ECNN framework.